`genQE_3N` — Three-Nucleon GCF Event Generator

Physics & Pipeline Documentation

GCF with 3N Project

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1 Overview

`genQE_3N` extends the two-nucleon GCF generator to **three-nucleon short-range correlations (3N SRC)**. It generates events of the form

$$e + A \longrightarrow e' + N_1 + N_2 + N_3 + (A{-}3)^*, \tag{1}$$

where the electron scatters off one nucleon of a correlated triplet. Only *ppn* triplets are generated. The default target is ^{12}C (residual ^{9}Be); `-A/-Z` select any nucleus with $A \geq 3$, $Z \geq 2$, $N \geq 1$ whose mass and $(A{-}3,\ Z{-}2)$ residual mass are in the generator's lookup table (`QEGenerator_3N::LookupNuclearMass`). For ^{3}He the triplet is the whole nucleus: there is no residual and no CM motion.

The key conceptual difference from the 2N generator is that *contact coefficients times universal functions* are replaced by **pre-computed 3D density tables** $\rho(\theta_{ab},\, p_a,\, p_b)$ built from Faddeev ^{3}He ground-state wave functions, used as a stand-in for the 3N SRC wave function in any nucleus.

2 From 2N Contacts to 3N Density Matrices

2.1 2N recap

In the 2N GCF the pair momentum distribution factorizes as

$$n^{(ij)}(\boldsymbol{k}) = C_{ij}^{A}\, |\varphi_{ij}(k)|^2, \tag{2}$$

separating a nucleus-dependent contact C_{ij}^{A} from a universal function φ_{ij}.

2.2 3N generalization

For three nucleons no analogous simple factorization exists. Instead, the full three-body correlation is encoded in a density matrix that depends on three independent kinematic variables describing the internal configuration of the triplet:

$$\rho_{3N}(\theta_{ab},\, p_a,\, p_b), \tag{3}$$

where, in the triplet rest frame,

- $p_a = |\boldsymbol{p}_1|$: momentum of the odd nucleon (in the tables, the neutron),
- $p_b = |\boldsymbol{p}_2 - \boldsymbol{p}_3|/2$: *half*-relative momentum of the other two (the *pp* pair),
- θ_{ab}: angle between $\boldsymbol{p}_a$ and $\boldsymbol{p}_b$.

The half-relative convention for p_b is the one the tables use: with it the mean kinetic energy of the AV18 table comes out at $\langle T \rangle = 45.6\,\mathrm{MeV}$, while reading the same table with the full relative momentum $\boldsymbol{p}_2 - \boldsymbol{p}_3$ gives 32 MeV. The code calls the two table axes k_{cm} $(= p_a)$ and k_{rel} $(= p_b)$.

The tables are plain-text files, loaded at initialization.

2.3 Data files

The interaction number `u` (second positional argument; `-u`/`--wave-function` in `gcf_generate`) selects the table:

u	File (in `array/`)
1	`he3_3Ndensity_AV18.dnst` (AV18)
2	`he3_3Ndensity_AV18UIX.dnst` (AV18 + Urbana IX)
3	`he3_3Ndensity_AV18TM.dnst` (AV18 + Tucson–Melbourne)
4	`he3_3Ndensity_CDBonn.dnst` (CD-Bonn)
5	`he3_3Ndensity_CDBonnTM.dnst` (CD-Bonn + Tucson–Melbourne)
6–10	`he3_3Ndensity_pD_*.dnst` (the same five, deuteron channel only)

The pD tables keep only the deuteron channels of the Faddeev wave function and share the grid, units and Jacobian of $u = 1$–5. Any other `u` aborts. The older `AV8/N2LO/G3/AV4_1D.dnst` tables are still in `array/` but are no longer loaded. The directory is the absolute path CMake bakes in as `GCF_3N_ARRAY_DIR`; the environment variable of the same name overrides it at run time.

Each file holds one value per line, $61 \times 101 \times 101 = 622{,}261$ lines, with θ_{ab} the outermost and p_b the innermost index:

Variable	Bins	Range
θ_{ab}	61	$[0°, 180°]$, $\Delta\theta = 3°$
$p_a/\hbar c$	101	$[0,\ 10]\,\mathrm{fm}^{-1}$, $\Delta k = 0.1\,\mathrm{fm}^{-1}$
$p_b/\hbar c$	101	$[0,\ 10]\,\mathrm{fm}^{-1}$, $\Delta k = 0.1\,\mathrm{fm}^{-1}$

Values are trilinearly interpolated at runtime; p_a or p_b above $10\,\mathrm{fm}^{-1}$ returns zero.

Normalization. A table stores the raw (Cartesian) density $|\tilde{\varphi}(p_a, p_b, \theta_{ab})|^2$ per unit $d^3p_a\, d^3p_b$, which is finite at $p_a = p_b = 0$. With momenta in GeV/c,

$$8\pi^2 \int \rho(\theta_{ab}, p_a, p_b)\, p_a^2\, p_b^2\, \sin\theta_{ab}\, dp_a\, dp_b\, d\theta_{ab} = \tfrac{1}{3} \tag{4}$$

(measured: 0.33308). The 1/3 is the isospin assignment: the table fixes which nucleon is the neutron, and the three equivalent choices sum to one. The $8\pi^2 = 4\pi \cdot 2\pi$ comes from the orientation of $\boldsymbol{p}_a$ and the azimuth of $\boldsymbol{p}_b$ about it. Powers of $\hbar c$ are absorbed into the contact C.

3 Three-Body Kinematics

3.1 Coordinate system

The three nucleon momenta are parameterized via two Jacobi-like vectors $\boldsymbol{p}_a$, $\boldsymbol{p}_b$ and a three-body center-of-mass momentum $\boldsymbol{P}_{\mathrm{CM}}$.

Internal frame. $\boldsymbol{p}_a$ is placed along $\hat{z}$ and $\boldsymbol{p}_b$ at polar angle θ_{ab} and azimuth ϕ_{baz} about it:

$$\boldsymbol{p}_a = (0,\ 0,\ p_a), \qquad \boldsymbol{p}_b = p_b\,(\sin\theta_{ab}\cos\phi_{baz},\ \sin\theta_{ab}\sin\phi_{baz},\ \cos\theta_{ab}). \tag{5}$$

Rotation to lab. Both vectors are then rotated by θ_a about $\hat{y}$ and by ϕ_a about $\hat{z}$, so $(\phi_a,\ \theta_a,\ \phi_{baz})$ act as three Euler angles, all sampled uniformly.

Individual momenta.

$$\boldsymbol{v}_1 = \boldsymbol{p}_a, \qquad \boldsymbol{v}_2 = -\tfrac{1}{2}\,\boldsymbol{p}_a + \boldsymbol{p}_b, \qquad \boldsymbol{v}_3 = -\tfrac{1}{2}\,\boldsymbol{p}_a - \boldsymbol{p}_b. \tag{6}$$

so $\boldsymbol{p}_b = (\boldsymbol{v}_2 - \boldsymbol{v}_3)/2$, matching the tables. The map from $(\boldsymbol{P}_{\mathrm{CM}}, \boldsymbol{p}_a, \boldsymbol{p}_b)$ to the three nucleon momenta has unit determinant, so $d^3p_1\, d^3p_2\, d^3p_3 = d^3P_{\mathrm{CM}}\, d^3p_a\, d^3p_b$. After Euler rotation, each nucleon receives an additional $\boldsymbol{P}_{\mathrm{CM}}/3$ boost. The $(A{-}3)$ residual recoils with $\boldsymbol{p}_{A-3} = -\boldsymbol{P}_{\mathrm{CM}}$.

3.2 Center-of-mass motion

$$P_{\mathrm{CM},\alpha} \sim \mathcal{N}(0,\ \sigma_{\mathrm{CM}}), \qquad \sigma_{\mathrm{CM}} = 0.15\,\mathrm{GeV}/c \text{ (default)}. \tag{7}$$

The default is the ^{12}C 2N-pair literature value. `-s` sets σ_{CM} directly. In `genQE_3N_FSI` (`-K`) and `gcf_generate` (`--kF`), giving a Fermi momentum k_F sets $\sigma_{\mathrm{CM}} = \sqrt{3/5}\,k_F$ unless σ_{CM} is also given, the CM width of three uncorrelated mean-field nucleons. `-C 0` (`--no-cm`) switches the CM motion off, and it is always off for $A = 3$.

4 Event Generation Pipeline

4.1 Step 1: Triplet type assignment

Currently only *ppn* triplets are generated. All three nucleons are initialized as protons, then exactly one — chosen uniformly at random among N_1, N_2, N_3 — is reassigned to a neutron. Thus the leading nucleon N_1 can be either a proton or a neutron (with probability 2/3 and 1/3 respectively). The weight carries a factor of 3 to account for the three equivalent assignments.

4.2 Step 2: Sample three-body CM momentum

$$\boldsymbol{P}_{\mathrm{CM}} \sim \mathcal{N}^3(0,\ \sigma_{\mathrm{CM}}). \tag{8}$$

The Gaussian is the CM distribution itself, so it adds no weight factor.

4.3 Step 3: Sample orientation (Euler angles)

$$\phi_a \in [-\pi,\ \pi], \qquad \cos\theta_a \in [-1,\ 1], \qquad \phi_{baz} \in [-\pi,\ \pi]. \tag{9}$$

Phase-space weight factor: $\Delta\phi_a\,\Delta(\cos\theta_a)\,\Delta\phi_{baz} = 8\pi^2$.

4.4 Step 4: Sample relative momenta

$$p_a \in [0,\ 5.0]\,\mathrm{GeV}/c, \qquad p_b \in [0,\ 5.0]\,\mathrm{GeV}/c, \qquad \theta_{ab} \in [10^{-4},\ \pi]. \tag{10}$$

Here p_b is the half-relative momentum of Sec. 2, the variable the tables are read in. Phase-space weight factor: $5.0 \times 5.0 \times (\pi - 10^{-4})$.

4.5 Step 5: Construct individual momenta

Build $\boldsymbol{v}_{1,2,3}$ from the Jacobi decomposition and apply the Euler rotation. The event is rejected ($w = 0$) unless every internal momentum satisfies

$$k_{\min} \le |\boldsymbol{v}_i| \le 5\,\mathrm{GeV}/c, \qquad i = 1, 2, 3, \tag{11}$$

with $k_{\min} = 0.25\,\mathrm{GeV}/c$ by default, or k_F when `-K` (`--kF`) is given: the SRC condition $k > k_F$ on the initial state. The cut acts before the CM boost. Then add $\boldsymbol{P}_{\mathrm{CM}}/3$ to each nucleon and set $\boldsymbol{p}_{A-3} = -\boldsymbol{P}_{\mathrm{CM}}$.

4.6 Step 6: Residual-nucleus energy

The $(A-3)$ residual (for ^{12}C, ^{9}Be) is on shell, with mass from the lookup table plus an excitation energy ($E^* = 0$ in the drivers):

$$m_{A-3} = M(A-3,\ Z-2) + E^*, \qquad E_{A-3} = \sqrt{|\boldsymbol{p}_{A-3}|^2 + m_{A-3}^2}. \tag{12}$$

For ^{3}He there is no residual and $E_{A-3} = 0$.

Spectator energies:

$$E_2 = \sqrt{|\boldsymbol{v}_2|^2 + m_N^2}, \qquad E_3 = \sqrt{|\boldsymbol{v}_3|^2 + m_N^2}. \tag{13}$$

Leading nucleon energy from conservation (off shell):

$$E_1 = m_A - E_{A-3} - E_2 - E_3. \tag{14}$$

4.7 Step 7: Electron scattering vertex

The scattered electron direction is sampled uniformly:

$$\cos\theta_k \in [\cos\theta_k^{\max}, \cos\theta_k^{\min}], \qquad \phi_k \in [-\pi, \pi], \tag{15}$$

with $[\theta_k^{\min}, \theta_k^{\max}] = [5^\circ, 40^\circ]$ in the drivers (`-t` in `genQE_3N_FSI`, `--theta-e` in `gcf_generate`). Phase-space weight factor: $\Delta(\cos\theta_k)\,\Delta\phi_k$.

Define

$$\boldsymbol{X} = \boldsymbol{v}_1 + \boldsymbol{p}_{\text{beam}}, \qquad Z = m_A + E_{\text{beam}} - E_2 - E_3 - E_{A-3}. \tag{16}$$

Requiring the struck nucleon to be on shell after absorbing the photon,

$$Z - E_k = \sqrt{m_N^2 + (\boldsymbol{X} - E_k\hat{k})^2}, \tag{17}$$

fixes the scattered electron energy:

$$E_k = \frac{Z^2 - m_N^2 - |\boldsymbol{X}|^2}{2\big(Z - \boldsymbol{X}\cdot\hat{k}\big)}. \tag{18}$$

The event is rejected if $E_k \le 0$ or $Z - E_k \le 0$, or if Q^2 falls outside the optional generation window (`-q` in `genQE_3N_FSI`, `--q2` in `gcf_generate`; open by default).

The virtual photon is $q = p_{\text{beam}} - k$ (four-vectors), and the lead nucleon receives the photon: $p_{\text{lead}} = (E_1, \boldsymbol{v}_1) + q$, so $E_{\text{lead}} = Z - E_k$.

4.8 Step 8: Jacobian

Integrating the energy-conservation δ-function over E_k produces the Jacobian

$$J_\delta = E_{\text{lead}} - \boldsymbol{p}_{\text{lead}}\cdot\hat{k}, \qquad w \;*=\; \left|\frac{E_{\text{lead}}}{J_\delta}\right|, \tag{19}$$

from $|\partial f/\partial E_k| = J_\delta/E_{\text{lead}}$, where the inner derivative of $(\boldsymbol{X} - E_k\hat{k})^2$ is $-2\,\boldsymbol{p}_{\text{lead}}\cdot\hat{k}$.

4.9 Step 9: Cross section and density weight

$$w \;*=\; \frac{1}{2}\,(2\pi)^{-7}\,C\,t \;\times\; \sigma_{eN}(E_{\text{beam}}, \boldsymbol{k}, \boldsymbol{p}_{\text{lead}}) \;\times\; \rho_{3N}(\theta_{ab}, p_a, p_b), \tag{20}$$

where $C = 0.12$ and $t = 2$ are normalization constants. σ_{eN} is the de Forest cross section of the 2N generator with Kelly form factors, cc1 by default (`-c onshell|cc1|cc2`).

4.10 Step 10: Density evaluation

The table always has the neutron as nucleon 1 and the pp pair as nucleons 2,3. The sampled triplet is therefore relabelled according to which of N_1, N_2, N_3 is the neutron, $(\boldsymbol{v}'_1, \boldsymbol{v}'_2, \boldsymbol{v}'_3)$, and the table is read at

$$p'_a = |\boldsymbol{v}'_1|, \qquad p'_b = \tfrac{1}{2}|\boldsymbol{v}'_2 - \boldsymbol{v}'_3|, \qquad \theta'_{ab} = \angle(\boldsymbol{v}'_1,\, \boldsymbol{v}'_2 - \boldsymbol{v}'_3). \tag{21}$$

The relabelling is a pointwise re-evaluation of the same density: the map between the two Jacobi sets has unit determinant, so it carries no Jacobian. Because the table is a density per $d^3p_a\, d^3p_b$ and the generator samples p_a, p_b and θ_{ab} uniformly, the weight needs the measure of the *sampled* variables, $d^3p_a\, d^3p_b = p_a^2\, p_b^2 \sin\theta_{ab}\; dp_a\, dp_b\, d\theta_{ab}\, d\Omega_a\, d\phi_{baz}$:

$$\rho_{3N} = p_a^2\, p_b^2\, \sin\theta_{ab} \times \text{interpolate}\big[\rho_{\text{table}}(\theta'_{ab},\, p'_a/\hbar c,\, p'_b/\hbar c)\big]. \tag{22}$$

The measure uses the unprimed (sampled) variables, the table the primed ones.

5 Weight Summary

Collecting all factors:

$$\begin{aligned} w = 3 \times 8\pi^2 \times 25\,(\pi - 10^{-4}) \times \Delta(\cos\theta_k)\, \Delta\phi_k \times \left|\frac{E_{\text{lead}}}{J_\delta}\right| \times \frac{C\,t}{2\,(2\pi)^7} \\ \times\, \sigma_{eN} \times p_a^2\, p_b^2 \sin\theta_{ab}\; \rho_{\text{table}}(\theta'_{ab}, p'_a, p'_b) \times \Theta(\text{cuts}), \end{aligned} \tag{23}$$

where Θ(cuts) stands for the internal-momentum window, the E_k conditions and the Q^2 window. Only events with $w > 0$ are written; `hAttempts` counts every attempt for per-attempt normalization.

5.1 Relation to `3N_Generator.pdf`

The code follows the tables where they disagree with the derivation in `3N_Generator.pdf`:

- Eq. (6) defines $\boldsymbol{p}_b = \boldsymbol{p}_2 - \boldsymbol{p}_3$; the tables, and the code, use $(\boldsymbol{p}_2 - \boldsymbol{p}_3)/2$. With the full relative momentum the map from $(\boldsymbol{p}_1, \boldsymbol{p}_2, \boldsymbol{p}_3)$ has $|\det| = 1/8$ instead of 1.
- Eq. (9) is inverted: the measure is $p_a^2\, p_b^2 \sin\theta_{ab}$, not its reciprocal.
- Eq. (25) leaves out $p_a^2\, p_b^2 \sin\theta_{ab}$, which the Cartesian tables require.
- Eq. (18) dots $\boldsymbol{p}'_1$ into the whole argument $\boldsymbol{p}_1 + \boldsymbol{p}_{\text{beam}} - \hat{k}$ instead of the inner derivative $-\hat{k}$; the code uses $J_\delta = E_{\text{lead}} - \boldsymbol{p}_{\text{lead}} \cdot \hat{k}$.

6 Alternative Momentum Parameterization

For analysis purposes (Dalitz/ternary diagrams, spectral functions), an alternative parameterization in terms of momentum fractions is available:

$$f_i = \frac{|\boldsymbol{p}_i|}{|\boldsymbol{p}_1| + |\boldsymbol{p}_2| + |\boldsymbol{p}_3|}, \qquad f_1 + f_2 + f_3 = 1. \tag{24}$$

Here $p_{\text{tot}} = p_1 + p_2 + p_3$ in the triplet rest frame, where $\boldsymbol{p}_1 + \boldsymbol{p}_2 + \boldsymbol{p}_3 = 0$. Integrating out the overall orientation, $d^3p_1\, d^3p_2 = 8\pi^2\, p_1\, p_2\, p_3\; dp_1\, dp_2\, dp_3$, and $dp_1\, dp_2\, dp_3 = p_{\text{tot}}^2\; dp_{\text{tot}}\, df_1\, df_2$, so the density per $dp_{\text{tot}}\, df_1\, df_2$ is

$$\frac{dN}{dp_{\text{tot}}\, df_1\, df_2} \;\propto\; p_1\, p_2\, p_3\, p_{\text{tot}}^2\; \rho_{\text{table}}, \tag{25}$$

up to the constant area factor of the ternary coordinates; it vanishes at the edges of the ternary plot. `QEGenerator_3N::get_rho_ptot_f1f2f3` currently uses the reciprocal, $J = (8/\sqrt{3})/(p_1\, p_2\, p_3\, p_{\rm tot}^2)$, which is wrong (issue #9). Event generation does not use it (it calls `get_rho`); only the ternary and θ-plot diagnostics (`Ternary.cc`, `analysis/theta_plot*.cpp`) do.

7 Differences from the 2N Generator

Aspect	`genQE` (2N)	`genQE_3N` (3N)
Correlated cluster	Pair (NN)	Triplet (NNN)
Structure input	Contacts $C_{ij}\,\varphi^2(k)$	^{3}He Faddeev density $\rho(\theta_{ab}, p_a, p_b)$
Residual system	$(A{-}2)^*$	$(A{-}3)^*$
Phase space dim.	6D (CM + rel)	9D (CM + Euler + Jacobi)
Electron vertex	Quadratic solve for $k_\perp$	Direct solve for E_k
$\sigma_{\rm CM}$	0.10–0.15 GeV/c	0.15 GeV/c, or $\sqrt{3/5}\,k_F$
Target nuclei	^{2}H through ^{208}Pb	^{12}C default; $A \geq 3$ in the mass table

8 Summary of Algorithm

For each Monte Carlo trial:

1. Assign triplet types (ppn); weight $\times\, 3$.
2. Sample $\boldsymbol{P}_{\rm CM}$ (Gaussian, $\sigma = 0.15\,\text{GeV}/c$ by default, or $\sqrt{3/5}\,k_F$; none for $A = 3$).
3. Sample Euler angles $(\phi_a, \theta_a, \phi_{baz})$ uniformly.
4. Sample Jacobi magnitudes (p_a, p_b) and angle θ_{ab} uniformly.
5. Construct $\boldsymbol{v}_{1,2,3}$ via Jacobi decomposition + Euler rotation; apply $k_{\rm min} \leq |\boldsymbol{v}_i| \leq 5\,\text{GeV}/c$; add the CM boost.
6. Compute residual and spectator energies; derive E_1 from conservation.
7. Sample electron direction (θ_k, ϕ_k); solve for E_k; apply the optional Q^2 window.
8. Compute Jacobian $J_\delta = E_{\rm lead} - \boldsymbol{p}_{\rm lead} \cdot \hat{k}$.
9. Evaluate σ_{eN} (cc1 by default, Kelly form factors).
10. Relabel so the neutron is nucleon 1, look up $\rho_{\rm table}$, and multiply by $p_a^2\, p_b^2 \sin\theta_{ab}$.
11. Multiply all weight factors; store event if $w > 0$.